MIL-S-19500/202A
2 August 1962
SUPERSEDING
MIL-S-19500/202
23 April 1962

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICES, DIODES, SILICON, POWER RECTIFIERS,

TYPES JAN-1N538M, JAN-1N540M, AND JAN-1N547M

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon (low power) rectifiers.

1.2 Ratings.

	Type	v_r	V_R	V_R(rms)	i_f	I_o (T_A = 150°C)	i_f (surge) (1/120 sec)	Barometric pressure
		Vdc	Vdc	Vac	ma	mAdc	a	mm Hg
Minimum	JAN-1N538M	---	---	---	---	---	---	15
	JAN-1N540M	---	---	---	---	---	---	15
	JAN-1N547M	---	---	---	---	---	---	15
Maximum	JAN-1N538M	240	200	140	750	250	15	---
	JAN-1N538M	480	400	280	750	250	15	---
	JAN-1N547M	720	600	420	750	250	15	---

OPERATING TEMPERATURE: -65° to + 150° C.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-S-19491 - Semiconductor Devices, Preparation for Delivery of.
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5960

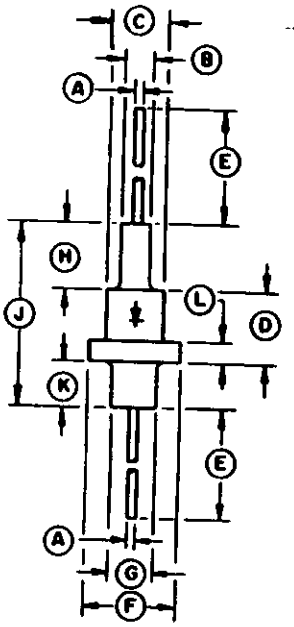
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3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with Specification MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in Specification MIL-S-19500, and as follows:

Vac ..... Variable voltage source.
IR_o ..... Reverse current, average



ALL DIMENSIONS IN INCHES

DIM.	MIN	MAX
A	0.017 DIA	0.033 DIA
B	---	0.125 DIA
C	0.240 DIA	0.280 DIA
D	---	0.330
E	1.000	---
F	0.310 DIA	0.400 DIA
G	---	0.180 DIA
H	---	0.400
J	---	0.800
K	---	0.200
L	---	0.100

Figure 1 - Semiconductor devices, diodes, types JAN-1N538M, JAN-1N540M, and JAN-1N547M.

3.3 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.4 Marking. The following markings specified in MIL-S-19500 shall be omitted from the body.

(a) Country of origin
(b) Acceptance date and inspection lot identification
(c) Manufacturer's identification
(d) Color band identification

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with Specification MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.2.1 Qualification sample sizes and acceptance numbers. For purposes of qualification inspection, the LTPDs listed in this detail specification shall govern. The sample sizes and acceptance numbers shall be as listed in tables II and III of specification MIL-S-19500.

4.3 Acceptance inspection. Acceptance inspection shall consist of groups A and B, and inspection of preparation for delivery (see 4.4).

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I and shall be on a sublot basis.

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4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II. Subgroups 1 through 7 shall be on a lot basis. Subgroup 8 shall be performed on the highest voltage unit in the lot. Subgroups 9 and 10 shall be on a sublot basis. The manufacturer has the option of performing all Group B inspection on a sublot basis.

Table I - Group A inspection.

Examination or test	Conditions 1/	LTPD	Minimum rejection No.	Symbol	Limits		Unit
					Min	Max	
Subgroup 1							
Visual and mechanical examination.	---	5%	4	---	---	---	---
Subgroup 2							
Forward voltage:	Method 4011						
JAN-1N538M	I_F = 0.5 Adc			V_F	---	1.1	Vdc
JAN-1N540M	I_F = 0.5 Adc			V_F	---	1.1	Vdc
JAN-1N547M	I_F = 0.5 Adc			V_F	---	1.2	Vdc
Reverse current:	Method 4016						
JAN-1N538M	V_R = 200 Vdc			I_R	---	10	uAdc
JAN-1N540M	V_R = 400 Vdc			I_R	---	10	uAdc
JAN-1N547M	V_R = 600 Vdc			I_R	---	10	uAdc
Reverse current at peak reverse voltage:	Method 4016						
JAN-1N538M	v_r = 240 vdc			I_R	---	100	uAdc
JAN-1N540M	v_r = 480 vdc			I_R	---	100	uAdc
JAN-1N547M	v_r = 720 vdc	5%	4	I_R	---	100	uAdc
Reverse current, average (See 4.5.1):	Method 4046 T_A = 150° C, min R_p = 22,000 ohms			---	---	---	---
JAN-1N538M	Vac = 140 Vac I_o = 250 mAdc			IR_o	---	350	uAdc
JAN-1N540M	Vac = 280 Vac I_o = 250 mAdc			IR_o	---	350	uAdc
JAN-1N547M	Vac = 420 Vac I_o = 250 mAdc			IR_o	---	350	uAdc
Reverse current:	Method 4016 T_A = 150°C, min						
JAN-1N538M	V_R = 200 Vdc			I_R	---	500	uAdc
JAN-1N540M	V_R = 400 Vdc			I_R	---	500	uAdc
JAN-1N547M	V_R = 600 Vdc			I_R	---	500	uAdc

1/ See MIL-STD-750

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Table II - Group B inspection.

Examination or test	Conditions 1/	LTPD	Minimum rejection No.	Symbol	Limits		Unit
					Min	Max	
Subgroup 1							
Physical dimensions	---	10%	5	---	---	---	---
Subgroup 2							
Low temperature:	T_A = -65°C, max						
Reverse current:	Method 4016						
JAN-1N538M	V_R = 200 Vdc			I_R	---	100	uAdc
JAN-1N540M	V_R = 400 Vdc			I_R	---	100	uAdc
JAN-1N547M	V_R = 600 Vdc			I_R	---	100	uAdc
Forward voltage	Method 4011						
	I_F = 0.5 Adc			V_F	---	1.3	Vdc
End points:							
Forward voltage:							
JAN-1N538M	I_F = 0.5 Adc			V_F	---	1.1	Vdc
JAN-1N540M	I_F = 0.5 Adc			V_F	---	1.1	Vdc
JAN-1N547M	I_F = 0.5 Adc			V_F	---	1.2	Vdc
Reverse current:							
JAN-1N538M	V_R = 200 Vdc	10%	5	IR	---	10	uAdc
JAN-1N540M	V_R = 400 Vdc			I_R	---	10	uAdc
JAN-1N547M	V_R = 600 Vdc			I_R	---	10	uAdc
Reverse current at peak reverse voltage:							
JAN-1N538M	v_r = 240 vdc			I_R	---	100	uAdc
JAN-1N540M	v_r = 480 vdc			I_R	---	100	uAdc
JAN-1N547M	v_r = 720 vdc			I_R	---	100	uAdc
Reverse current:	T_A = 150°C, min						
JAN-1N538M	V_R = 200 Vdc			I_R	---	500	uAdc
JAN-1N540M	V_R = 400 Vdc			I_R	---	500	uAdc
JAN-1N547M	V_R = 600 Vdc			I_R	---	500	uAdc
Subgroup 3 2/							
Temperature cycling	Method 1051 test condition B, except -65° + 0°C to + 175° C -0°C			---	---	---	---
Thermal shock (glass strain) 2/.	Method 1056, test cond. B	10%	5	---	---	---	---
Moisture resistance 3/, 4/	Method 1021			---	---	---	---
End points: (Same as for subgroup 2)							

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Table II - Group B inspection (Cont'd).

Examination or test	Conditions 1/	LTPD	Minimum rejection No.	Symbol	Limits		Unit
					Min	Max	
Subgroup 4							
Salt atmosphere (corrosion) 2/	Method 1041	10%	5	---	---	---	---
End points: (Same as for subgroup 2)	---			---	---	---	---
Subgroup 5							
Shock 2/	Method 2016 1500 G; 5 blows each in orientations X_1, Y_1, and Y_2 (total of 15 blows)	10%	5	---	---	---	---
Vibration fatigue 2/	Method 2046			---	---	---	---
Vibration, variable frequency.	Method 2056			---	---	---	---
Constant acceleration	Method 2006; 10,000 G;			---	---	---	---
End points: (Same as for subgroup 2)	---			---	---	---	---
Subgroup 6							
Surge current:	Method 4066: T_A = 150° C; 10 surges, 1/120 sec. duration: i_f (surge) = 15a.	10%	5				
JAN-1N538M	Vac = 140 Vac I_o = 250 mAdc			---	---	---	
JAN-1N540M	Vac = 280 Vac I_o = 250 mAdc			---	---	---	
JAN-1N547M	Vac = 420 Vac I_o = 250 mAdc			---	---	---	
End points: (Same as for subgroup 2)							
Subgroup 7							
Terminal strength (tension test) 2/	Method 2036: test cond. A: 2 lb: 1 minute	10%	5	---	---	---	---
Terminal strength (lead fatigue) 2/	Method 2036: test cond. E 4 arcs each lead						
End points: (Same as for subgroup 2)	---			---	---	---	---

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Table II - Group B inspection (Cont'd).

Examination or test	Conditions 1/	LTPD	Minimum rejection No.	Symbol	Limits		Unit
					Min	Max	
Subgroup 8							
Barometric pressure reduced (altitude operation):	Method 1001: 15 mm Hg t = 1 min	10	5				
JAN-1N538M	V_R = 200 Vdc			I_R	---	10	uAdc
JAN-1N540M	V_R = 400 Vdc			I_R	---	10	uAdc
JAN-1N547M	V_R = 600 Vdc			I_R	---	10	uAdc
End points: (Same as for subgroup 2)	---			---	---	---	---
Subgroup 9							
Steady state operation life	Method 1026 T_A = 150°C, min	λ = 5	4				
JAN-1N538M	V_{ac} = 140 Vac I_o = 250 mAdc	---	---	---	---	---	---
JAN-1N540M	V_{ac} = 280 Vac I_o = 250 mAdc	---	---	---	---	---	---
JAN-1N547M	V_{ac} = 420 Vac I_o = 250 mAdc	---	---	---	---	---	---
End points:							
Forward voltage	T_A = 25°C, min						
JAN-1N538M	I_F = 0.5 Adc	---	---	V_F	---	1.1	Vdc
JAN-1N540M	I_F = 0.5 Adc	---	---	V_F	---	1.1	Vdc
JAN-1N547M	I_F = 0.5 Adc	---	---	V_F	---	1.2	Vdc
Reverse current:	T_A = 25°C, min						
JAN-1N538M	V_R = 200 Vdc	---	---	I_R	---	10	uAdc
JAN-1N540M	V_R = 400 Vdc	---	---	I_R	---	10	uAdc
JAN-1N547M	V_R = 600 Vdc	---	---	I_R	---	10	uAdc
Subgroup 10							
High temperature life (non-operating)	Method 1031 T_A = 175°C, min	λ = 5	4	---	---	---	---
End points: (Same as for subgroup 9)	---	---	---	---	---	---	---

1/ See MIL-STD-750.
2/ Destructive tests.
3/ No lead-fatigue test prior to cycling. The vibration requirement shall apply.
4/ For qualification inspection, the moisture resistance test shall be performed with the working voltage, V_R, applied and shall be monitored once during each of the last 5 cycles.

4.4 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with Specification MIL-S-19491 to verify conformance with requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 The notes specified in Specification MIL-S-19500 are applicable to their specification.

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